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CE005

LAB 8

Aim: Creating Web API in .NET core

Create .NET core Web API project to achieve the following:

- **Define Student model with appropriate fields.**
- **Define corresponding Controller with GET, POST, PUT and DELETE methods.**
- **Test API endpoints using Swagger / Postman**

Student.cs

```
namespace Lab8.Models
{
    public class Student
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string Email { get; set; }
        public string Course { get; set; }
        public int Age { get; set; }
    }
}
```

StudentController.cs

```
using Lab8.Models;
using Microsoft.AspNetCore.Http;
using Microsoft.AspNetCore.Mvc;

namespace Lab8.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentController : ControllerBase
    {
        public static readonly List<Student> _students = new List<Student> {
            new Student { Id = 1, Name = "Vidhi", Email = "vidhi@gmail.com", Course = "CE", Age =
21 },

```

```

        new Student { Id = 2, Name = "Palak", Email = "palak@gmail.com", Course = "CE", Age
= 20 },
        new Student { Id = 3, Name = "Harmi", Email = "harmi@gmail.com", Course = "CE", Age
= 21 },
        new Student { Id = 4, Name = "Teju", Email = "teju@gmail.com", Course = "CE", Age =
20 },
    };

```

```

[HttpGet]
public ActionResult GetAllStudents()
{
    return Ok(_students);
}

```

```

[HttpGet("{id}")]
public ActionResult GetStudent(int id)
{
    var student = _students.FirstOrDefault(s => s.Id == id);

    if(student == null)
        return NotFound();

    return Ok(student);
}

```

```

[HttpPost]
public ActionResult PostStudent(Student student)
{
    _students.Add(student);
    return Ok();
}

```

```

[HttpPut("{id}")]
public ActionResult PutStudent(int id, Student student)
{
    if(id != student.Id)
        return BadRequest();

    var existingStudent = _students.FirstOrDefault(s => s.Id == id);

    if(existingStudent == null)
        return NotFound();

    existingStudent.Id = id;
}

```

```

        existingStudent.Name = student.Name;
        existingStudent.Email = student.Email;
        existingStudent.Age = student.Age;
        existingStudent.Course = student.Course;

        return NoContent();
    }

    [HttpDelete("{id}")]
    public ActionResult DeleteStudent(int id)
    {
        var student = _students.FirstOrDefault(s => s.Id == id);

        if( student == null)
            return NotFound();

        _students.Remove(student);
        return NoContent();
    }
}

```

GET ALL

Student

GET /api/Student

Parameters

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
  https://localhost:7126/api/Student \
  -H 'accept: */*'
```

Request URL

```
https://localhost:7126/api/Student
```

Server response

CodeDetails

200

Response body

```
{
  "id": 1,
  "name": "Vidhi",
  "email": "vidhi@gmail.com",
  "course": "CE",
  "age": 21
},
{
  "id": 2,
  "name": "Palak",
  "email": "palak@gmail.com",
  "course": "CE",
  "age": 20
},
{
  "id": 3,
  "name": "Harni",
  "email": "harni@gmail.com",
  "course": "CE",
  "age": 21
},
{
  "id": 4,
  "name": "Teju",
  "email": "teju@gmail.com",
  "course": "CE",
  "age": 20
},
{
  "id": 5,
  "name": "Teju",
  "email": "teju@gmail.com",
  "course": "CE",
  "age": 20
}
}
```

Response headers

```
content-type: application/json; charset=utf-8
date: Tue, 10 Feb 2026 03:31:12 GMT
server: Kestrel
```

Responses

Code	Description	Links
200	OK	No links

GET https://localhost:7126/api/Student

Send

Params Authorization Headers (6) Body Pre-request Script Tests Settings Cookies

Query Params

Key	Value	Bulk Edit
Key	Value	

Body Cookies Headers (4) Test Results Status: 200 OK Time: 21 ms Size: 439 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   {
3     "id": 1,
4     "name": "Vidhi",
5     "email": "vidhi@gmail.com",
6     "course": "CE",
7     "age": 21
8   },
9   {
10    "id": 2,
11    "name": "Palak",
12    "email": "palak@gmail.com",
13    "course": "CE",
14    "age": 20
15  },
16  {
17    "id": 3,
18    "name": "Harmi",
19    "email": "harmi@gmail.com",
20    "course": "CE",
21    "age": 21
22  },
23  {
24    "id": 4,
25    "name": "Teju",
26    "email": "teju@gmail.com",
27    "course": "CE",
28    "age": 20
29  }
30 }
```

GET https://localhost:7126/api/Student/1

Send

Params Authorization Headers (6) Body Pre-request Script Tests Settings Cookies

Query Params

Key	Value	Bulk Edit
Key	Value	

Body Cookies Headers (4) Test Results Status: 200 OK Time: 21 ms Size: 220 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "id": 1,
3   "name": "Vidhi",
4   "email": "vidhi@gmail.com",
5   "course": "CE",
6   "age": 21
7 }
```

GET

GET /api/Student/{id}

Parameters

Cancel

Name	Description
id	required integer(\$int32) (path)

3

Execute

Clear

Responses

Curl

curl -X 'GET' \n'https://localhost:7126/api/Student/3' \n-H 'accept: */*'

Request URL

https://localhost:7126/api/Student/3

Server response

Code	Details
200	Response body {\n "id": 3,\n "name": "Harml",\n "email": "harml@gmail.com",\n "course": "c#",\n "age": 21\n} Response headers content-type: application/json; charset=utf-8\ndate: Tue, 10 Feb 2026 03:29:04 GMT\nserver: Kestrel

Responses

Code	Description	Links
200	OK	No links

POST

POST /api/Student

Parameters

Cancel

Reset

No parameters

Request body

application/json

```
{  "id": 5,  "name": "Janvi",  "email": "janvi@gmail.com",  "course": "IT",  "age": 21}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \  'https://localhost:7126/api/Student' \  -H 'accept: */*' \  -H 'Content-Type: application/json' \  -d '{  "id": 5,  "name": "Janvi",  "email": "janvi@gmail.com",  "course": "IT",  "age": 21  }'
```

Request URL

https://localhost:7126/api/Student

Server response

CodeDetails

200

Response headers

content-length: 0 date: Tue, 18 Feb 2026 03:29:55 GMT server: Kestrel

Responses

Code	Description	Links
200	OK	No links

PUT

PUT

/api/Student/{id}

Cancel

Reset

Name	Description
id * required	

integer(\$int32)
(path)

5

Request body

application/json

```
{
  "id": 5,
  "name": "Priya",
  "email": "priya@gmail.com",
  "course": "IT",
  "age": 21
}
```

Execute

Clear

Responses

Curl

```
curl -X 'PUT' \
  https://localhost:7126/api/Student/5 \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "id": 5,
    "name": "Priya",
    "email": "priya@gmail.com",
    "course": "IT",
    "age": 21
  }'
```

Request URL

https://localhost:7126/api/Student/5

Server response

Code	Details
204	

Undocumented

Response headers

date: Tue, 10 Feb 2026 03:33:16 GMT
server: Kestrel

Responses

Code	Description	Links
200	OK	No links

DELETE

DELETE

/api/Student/{id}

⌵

Parameters

Cancel

Name	Description
id * required	
integer(\$int32)	
(path)	

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \
  'https://localhost:7126/api/Student/5' \
  -H 'accept: */*'
```

Request URL

https://localhost:7126/api/Student/5

Server response

Code	Details
204	
Undocumented	
	Response headers <pre>date: Tue, 10 Feb 2026 03:33:27 GMT server: Kestrel</pre>

Responses

Code	Description	Links
200	OK	No links

Implement Model+Controller for any other entity of your choice for the University Management system.

[Classroom.cs](#)

```
namespace UniversityApi.Models
{
    public class Classroom
    {
        public int Id { get; set; }
        public string RoomNumber { get; set; } = string.Empty;
        public string BuildingName { get; set; } = string.Empty;
        public int Capacity { get; set; }
        public bool HasProjector { get; set; }
    }
}
```

[ClassroomsController.cs](#)

```
using Microsoft.AspNetCore.Mvc;
using UniversityApi.Models;

namespace UniversityApi.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class ClassroomsController : ControllerBase
    {
        private static readonly List<Classroom> Classrooms = new()
        {
            new Classroom
            {
                Id = 1,
                RoomNumber = "A101",
                BuildingName = "Main Block",
                Capacity = 60,
                HasProjector = true
            },
            new Classroom
            {
                Id = 2,
                RoomNumber = "B204",
```

```

        BuildingName = "Science Block",
        Capacity = 40,
        HasProjector = false
    }
};

[HttpGet]
public IActionResult GetAllClassrooms()
{
    return Ok(Classrooms);
}

[HttpGet("{id}")]
public IActionResult GetClassroomById(int id)
{
    var classroom = Classrooms.FirstOrDefault(c => c.Id == id);
    if (classroom == null)
        return NotFound();

    return Ok(classroom);
}

[HttpPost]
public IActionResult AddClassroom(Classroom classroom)
{
    classroom.Id = Classrooms.Max(c => c.Id) + 1;
    Classrooms.Add(classroom);

    return CreatedAtAction(nameof(GetClassroomById),
        new { id = classroom.Id }, classroom);
}

[HttpPut("{id}")]
public IActionResult UpdateClassroom(int id, Classroom updatedClassroom)
{
    var classroom = Classrooms.FirstOrDefault(c => c.Id == id);
    if (classroom == null)
        return NotFound();

    classroom.RoomNumber = updatedClassroom.RoomNumber;

```

```
classroom.BuildingName = updatedClassroom.BuildingName;  
classroom.Capacity = updatedClassroom.Capacity;  
classroom.HasProjector = updatedClassroom.HasProjector;
```

```
    return NoContent();  
}
```

```
[HttpDelete("{id}")]
```

```
public IActionResult DeleteClassroom(int id)
```

```
{  
    var classroom = Classrooms.FirstOrDefault(c => c.Id == id);  
    if (classroom == null)  
        return NotFound();
```

```
    Classrooms.Remove(classroom);  
    return NoContent();
```

```
}
```

```
}
```

```
}
```

GET ALL

Classrooms

GET /api/Classrooms

Parameters

No parameters

Execute Clear

Responses

Curl

curl -X 'GET' \ 'https://localhost:7151/api/Classrooms' \ -H 'accept: */*'

Request URL

https://localhost:7151/api/Classrooms

Server response

Code

Details

200

Response body

{ "id": 1, "roomNumber": "A101", "buildingName": "Main Block", "capacity": 60, "hasProjector": true }, { "id": 2, "roomNumber": "B204", "buildingName": "Science Block", "capacity": 40, "hasProjector": false }

Response headers

content-type: application/json; charset=utf-8 date: Tue, 10 Feb 2026 04:18:47 GMT server: Kestrel

Responses

Code	Description	Links
200	OK	No links

GET

GET

/api/Classrooms/{id}

⌵

Parameters

Cancel

Name	Description
id * required	
integer(int32)	
(path)	

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:7151/api/Classrooms/2' \
  -H 'accept: */*'
```

Request URL

```
https://localhost:7151/api/Classrooms/2
```

Server response

Code	Details
200	Response body <pre>{ "id": 2, "roomNumber": "B204", "buildingName": "Science Block", "capacity": 40, "hasProjector": false }</pre> Download Response headers <pre>content-type: application/json; charset=utf-8 date: Tue, 10 Feb 2020 04:28:02 GMT server: Kestrel</pre>

Responses

Code	Description	Links
200	OK	No links

POST

POST /api/Classrooms

Parameters

Cancel

Reset

No parameters

Request body

application/json

```
{  "id": 3,  "roomNumber": "34",  "buildingName": "narayan bhavan",  "capacity": 30,  "hasProjector": true}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \  'https://localhost:7151/api/Classrooms' \  -H 'accept: */*' \  -H 'Content-Type: application/json' \  -d '{  "id": 3,  "roomNumber": "34",  "buildingName": "narayan bhavan",  "capacity": 30,  "hasProjector": true}'
```

Request URL

https://localhost:7151/api/Classrooms

Server response

Code

Details

201

Undocumented

Response body

```
{  "id": 3,  "roomNumber": "34",  "buildingName": "narayan bhavan",  "capacity": 30,  "hasProjector": true}
```

Download

Response headers

```
content-type: application/json; charset=utf-8  date: Tue, 19 Feb 2024 04:21:59 GMT  location: https://localhost:7151/api/Classrooms/3  server: Kestrel
```

Responsees

Code	Description	Links
200	OK	No links

PUT

PUT

/api/Classrooms/{id}

⌵

Parameters

Cancel

Reset

Name	Description
id required	
integer(\$int32)	
(path)	3

Request body

application/json

```
{
  "id": 3,
  "roomNumber": "room2",
  "buildingName": "library",
  "capacity": 70,
  "hasProjector": true
}
```

Execute

Clear

Responses

Curl

```
curl -X 'PUT' \
  'https://localhost:7151/api/Classrooms/3' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "id": 3,
    "roomNumber": "room2",
    "buildingName": "library",
    "capacity": 70,
    "hasProjector": true
  }'
```

Request URL

https://localhost:7151/api/Classrooms/3

Server response

Code	Details
204	

Undocumented

Response headers

date: Tue, 10 Feb 2026 04:23:31 GMT
server: Kestrel

Responses

Code	Description	Links
200	OK	No links

DELETE

DELETE /api/Classrooms/{id}

Parameters

Cancel

Name	Description
id * required	
Integer(\$int32)	1
(path)	

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \
  https://localhost:7151/api/Classrooms/1' \
  -H 'accept: */*' 
```

Request URL

https://localhost:7151/api/Classrooms/1

Server response

Code	Details
204	
undocumented	Response headers <pre>date: Tue, 10 Feb 2026 04:24:31 GMT server: Kestrel</pre>

Responses

Code	Description	Links
200	OK	No links

ToDo API:

ToDoItem.cs

```
namespace ToDo.Models
{
    public class ToDoItem
    {
        public long Id { get; set; }
        public string? Name { get; set; }
        public bool IsComplete { get; set; }
    }
}
```

ToDoContext.cs

```
using System.Collections.Generic;
using Microsoft.EntityFrameworkCore;

namespace ToDo.Models
{
    public class TodoContext : DbContext
    {
        public TodoContext(DbContextOptions<TodoContext> options) : base(options)
        {
        }
        public DbSet<ToDoItem> TodoItems { get; set; } = null!;
    }
}
```

ToDoItemsController.cs

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.EntityFrameworkCore;
using ToDo.Models;

namespace TodoApi.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class TodoItemsController : ControllerBase
    {
        private readonly TodoContext _context;
```

```

public TodoItemsController(TodoContext context)
{
    _context = context;
}

[HttpGet]
public async Task<ActionResult<IEnumerable<ToDoItem>>> GetTodoItems()
{
    return await _context.TodoItems.ToListAsync();
}

[HttpGet("{id}")]
public async Task<ActionResult<ToDoItem>> GetTodoItem(long id)
{
    var todoItem = await _context.TodoItems.FindAsync(id);

    if (todoItem == null)
    {
        return NotFound();
    }

    return todoItem;
}

[HttpPost]
public async Task<ActionResult<ToDoItem>> PostTodoItem(ToDoItem todoItem)
{
    _context.TodoItems.Add(todoItem);
    await _context.SaveChangesAsync();

    return CreatedAtAction(nameof(GetTodoItem),
        new { id = todoItem.Id }, todoItem);
}

[HttpPut("{id}")]
public async Task<ActionResult> PutTodoItem(long id, ToDoItem todoItem)
{
    if (id != todoItem.Id)
    {
        return BadRequest();
    }

    _context.Entry(todoItem).State = EntityState.Modified;
}

```

```

        try
        {
            await _context.SaveChangesAsync();
        }
        catch (DbUpdateConcurrencyException)
        {
            if (!TodoItemExists(id))
            {
                return NotFound();
            }
            else
            {
                throw;
            }
        }

        return NoContent();
    }

    [HttpDelete("{id}")]
    public async Task<IActionResult> DeleteTodoItem(long id)
    {
        var todoItem = await _context.TodoItems.FindAsync(id);

        if (todoItem == null)
        {
            return NotFound();
        }

        _context.TodoItems.Remove(todoItem);
        await _context.SaveChangesAsync();

        return NoContent();
    }

    private bool TodoItemExists(long id)
    {
        return _context.TodoItems.Any(e => e.Id == id);
    }
}

```